

Genova Mongalo

913-260-8912 | Genova.Mongalo@gmail.com | linkedin.com/in/GMongalo | gmango.dev

SUMMARY

Georgia Tech M.S. Computer Science (AI) student and IEEE-published ML researcher. I compare models head to head, build evaluation harnesses, catch answers that sound right but are wrong, and write up why. I ship a clinical LLM assistant that cites PubMed, and I design RL reward functions, the incentive problem behind preference data.

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science, Artificial Intelligence emphasis

Atlanta, GA

Aug 2025 – Expected 2027

University of Missouri–Kansas City (UMKC)

B.S. in Computer Science, summa cum laude, GPA 3.82

Kansas City, MO

Aug 2021 – Dec 2024

Graduate coursework: Deep Learning, Machine Learning, Reinforcement Learning, Artificial Intelligence, Computer Vision, Bayesian Statistics (in progress).

EXPERIENCE

Founding AI/ML Product Engineer

Sep 2025 – Present

Geometry Health and Wellness, Kansas City, MO

- Built a context-aware **clinical LLM assistant** that answers from a patient's recent biomarkers and retrieved **PubMed abstracts with citations**, so guidance rests on published evidence instead of invented claims.
- Wrote the prompt construction and model routing across a **self-hosted 16B model** (Ollama) and cloud LLM providers, with **under 1 s** streaming latency.
- Ported clinically validated mobility assessments from research prototypes to a production **iOS app (Swift)**, matching the clinical reference within a fraction of a percent.

Federal AI/ML Engineer Contractor

May 2025 – Aug 2025

AFRL Sensors Directorate Internship Program, University of Dayton, Dayton, OH

- Devised a **game-theoretic approach to GAN training**; its synthetic data raised a downstream CNN classifier **50 percentage points** over baseline through transfer learning.
- Adapted state-of-the-art **PyTorch** classifiers to new imaging modalities, ran them in parallel on **Slurm**, and documented methods and results for cross-team review.

AI/ML Engineer Intern

Jun 2024 – Dec 2024

NSF REU, AI-Empowered Cybersecurity, UMKC, Kansas City, MO

- Built an **LLM-based ransomware detector** for industrial control systems from Windows API-call features: **99% binary / 91% family** accuracy as published.
- Solo author (with mentor guidance) of a paper accepted and presented at **IEEE Big Data 2024**.

Research Intern

May 2023 – Jul 2023

NSF REU, Consumer Networking, University of Missouri, Columbia, MO

- Built a **transformer** model predicting cybersickness severity on a 1–10 scale at 85% accuracy; the lab's work became a co-authored **IEEE TDSC 2025** paper on adversarial attacks against detection models.

Research Intern

Dec 2021 – Apr 2022

NASA Missouri Space Grant Consortium, Kansas City, MO

RESEARCH & PROJECTS

- Dysarthric Speech Recognition Evaluation (PyTorch, Whisper, wav2vec2)**. Georgia Tech team project; I built the transcription evaluation harness and ran the 43-fold leave-one-speaker-out study. Error grew about 13x on impaired speakers, and the models failed differently: Whisper wrote fluent, wrong English; wav2vec2 wrote visibly broken text, the safer failure for a nurse reading it.
- Ransomware Model Comparison (PyTorch, scikit-learn)**. Compared RoBERTa/ALBERT-style transformers and CNNs to baselines: 98.4% binary; traced weak 12-family results to class scarcity.
- Portfolio LLM Chatbot (gmango.dev)**. Iterated the system prompt against test questions and wrote a script that tests each model in a Groq/OpenRouter failover chain.
- AWS DeepRacer Reward Shaping (TD3)**. Hand-built a reward (centerline, heading, speed vs. steering) designed so the agent cannot score well by driving smoothly without finishing laps.

PUBLICATIONS

- R. K. Kundu, M. Denton, **G. Mongalo**, P. Calyam, K. A. Hoque, "Securing Virtual Reality Experiences: Unveiling and Tackling Cybersickness Attacks With Explainable AI," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 6, pp. 6040–6057, 2025.
- G. Mongalo**, "LLM Based Approach to Real Time Ransomware Detection for Industrial Control Systems," *IEEE Big Data*, 2024 (accepted and presented).

SKILLS

Model evaluation: model comparison, evaluation harnesses, error and failure-mode analysis, citation-grounded LLM output, prompt engineering, reward design, accuracy / MAE / weighted kappa, LaTeX and Markdown technical writing

Programming: Python, C/C++, Java/C#, TypeScript, Swift, SQL, Solidity, Racket/LISP **Spoken:** English, Spanish, Amharic

ML & tools: PyTorch, Hugging Face Transformers, scikit-learn, pandas/NumPy, Weights & Biases, Git, Docker, Slurm, AWS, GCP, Flask/FastAPI, React, Node.js